

Finite Element Analysis of a Thick-Walled Cylinder under Internal Pressure

1. PROBLEM DEFINITION

A long thick-walled cylindrical pressure vessel with open ends was analyzed under uniform internal pressure. The cylinder is assumed to be homogeneous, isotropic, and linearly elastic. Since the cylinder is sufficiently long, the radial displacement and stress distribution at the mid-length section can be compared with the classical Lamé solution for thick-walled cylinders.

The analysis was performed by using a two-dimensional axisymmetric finite element model. In the axisymmetric model, the radial direction was represented by the global X-axis and the axial direction was represented by the global Y-axis.

The assigned geometric and loading parameters used in this analysis are given in Table 1.

Parameter	Symbol	Value
Inner radius	r_i	58 mm
Outer radius	r_o	113 mm
Wall thickness	$t = r_o - r_i$	55 mm
Cylinder length	L	1380 mm
Mid-length section	$L/2$	690 mm
Internal pressure	P_i	53 MPa
Young's modulus	E	200000 MPa
Poisson's ratio	ν	0.30

Table 1. Problem Parameters

2. FINITE ELEMENT MODEL

The finite element model was created in ANSYS Mechanical APDL by using a 2D axisymmetric idealization. The rectangular model domain represents the r - z plane of the cylinder wall. The model extends from $r = 58$ mm to $r = 113$ mm in the radial direction and from $z = 0$ mm to $z = 1380$ mm in the axial direction.

The element type used in the analysis was PLANE182 with axisymmetric option. The material was defined as linear elastic, homogeneous, and isotropic with $E = 200000$ MPa and $\nu = 0.30$

In the axisymmetric model, the result components were interpreted as follows:

ANSYS result component	Physical meaning
UX	Radial displacement, u_r
UY	Axial displacement, u_z
SX	Radial stress, σ_r
SY	Axial stress, σ_z
SZ	Hoop stress, σ_θ

The internal pressure of 53 MPa was applied on the inner cylindrical surface at $r=r_i$. The outer cylindrical surface at $r=r_o$ was left traction-free. Since the cylinder has open ends, no axial constraints were applied to the end surfaces. To prevent rigid-body motion in the axial direction without over-constraining the model, only one node at the mid-length section was constrained in the axial direction:

$$U_Y = 0 \text{ at one node located at } r = 58 \text{ mm } z = 690 \text{ mm}$$

This constraint removes the rigid-body mode while preserving the open-ended behavior of the cylinder.

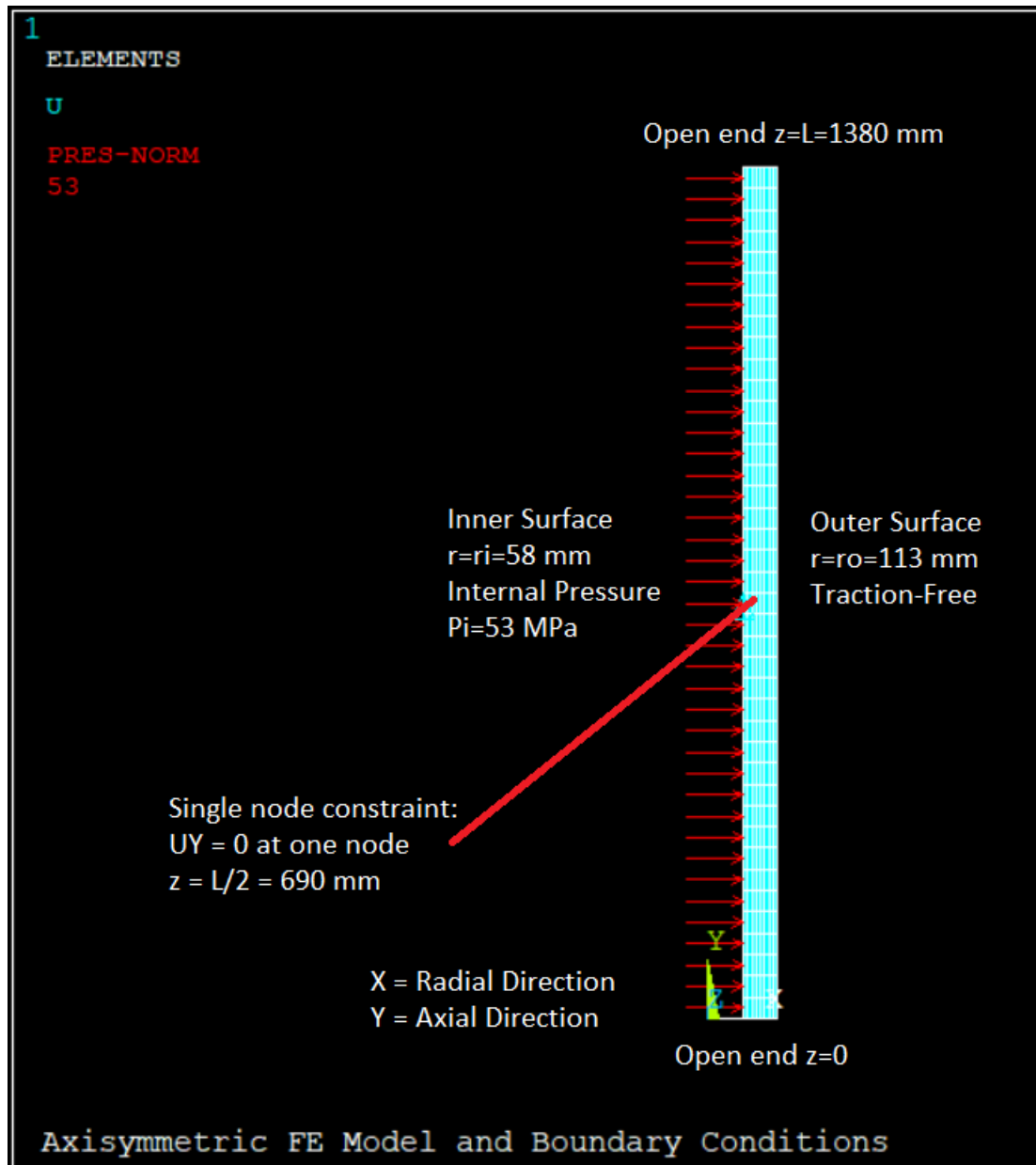


Figure 1 Two-dimensional axisymmetric finite element model and boundary conditions

3. MESH REFINEMENT STUDY

A mesh convergence study was performed by using three progressively refined mapped meshes: coarse, medium, and fine. The geometry, material properties, pressure loading, and boundary conditions were kept identical for all meshes. Only the number of elements was changed.

Mesh	Radial divisions	Axial divisions	Number of elements	Number of nodes
Coarse	8	40	320	369
Medium	16	80	1280	1377
Fine	30	200	6000	6231

Table 2. Mesh Properties

The radial displacement results at the inner and outer surfaces are summarized in Table 3. The values were obtained from the UX displacement component, which corresponds to radial displacement in the axisymmetric model.

Mesh	ur(ri) at z = L/2 (mm)	Error (%)	ur(ro) at z = L/2 (mm)	Error (%)	Maximum ur (mm)	Location of max. ur
Coarse	0,030983	0,022	0,021403	0,087	0,030983	r = 58 mm
Medium	0,030981	0,016	0,021417	0,021	0,030981	r = 58 mm
Fine	0,030979	0,009	0,021420	0,007	0,030979	r = 58 mm
Analytical	0,030976	—	0,021422	—	0,030976	r = 58 mm

Table 3. Mesh Convergence of Radial Displacement

The radial displacement results converge very rapidly. Even the coarse mesh gives displacement values very close to the analytical Lamé solution. The maximum radial displacement occurs at the inner surface, which is physically expected because the internal pressure acts directly on this surface.

Stress results are more sensitive to mesh refinement because stresses are calculated from displacement gradients. This can be seen from the radial stress and hoop stress contour values summarized in Table 4.

Mesh	σ_r min. (MPa)	σ_r max. (MPa)	σ_θ min. (MPa)	σ_θ max. (MPa)
Coarse	-47,7853	-0,7977	38,2658	88,2511
Medium	-50,2948	-0,3893	38,0980	89,5641
Fine	-51,5391	-0,2042	38,0143	90,1911
Analytical	-53,0000	0,0000	37,9143	90,9143

Table 4. Mesh Refinement Effect on Stress Results

As the mesh is refined, the radial stress at the inner surface approaches -53 MPa, and the radial stress at the outer surface approaches zero. Similarly, the hoop stress distribution approaches the analytical Lamé solution. The fine mesh was therefore used for the detailed path extraction and analytical comparison.

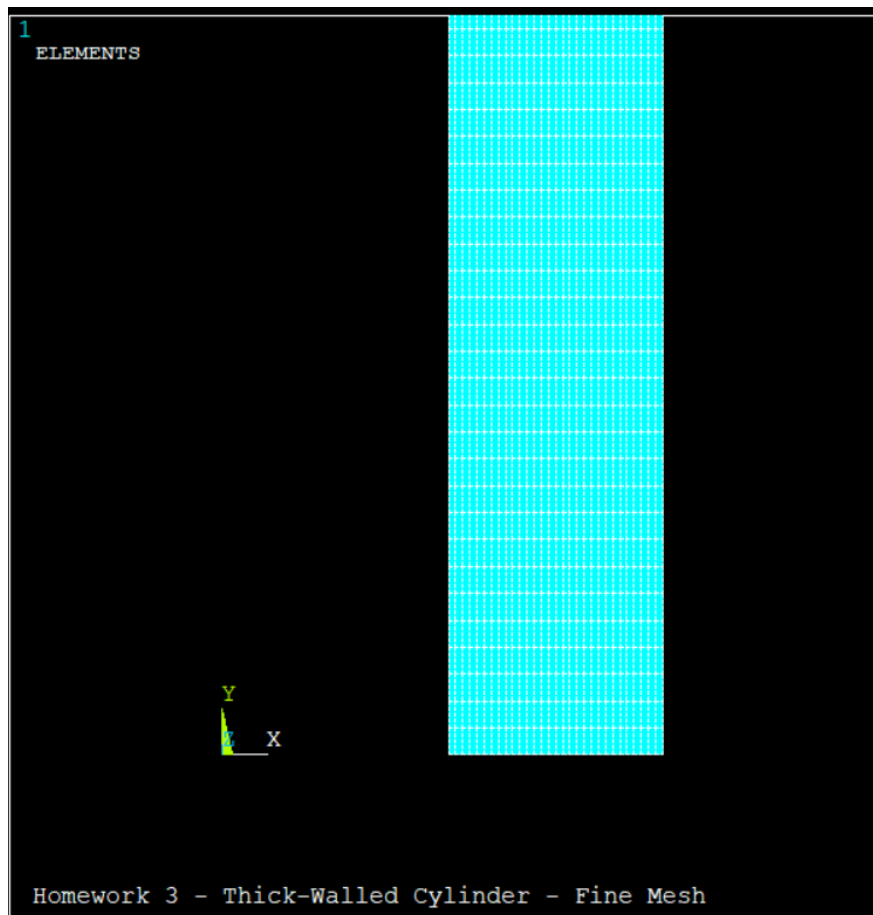


Figure 2 Fine mesh detail of the axisymmetric finite element model

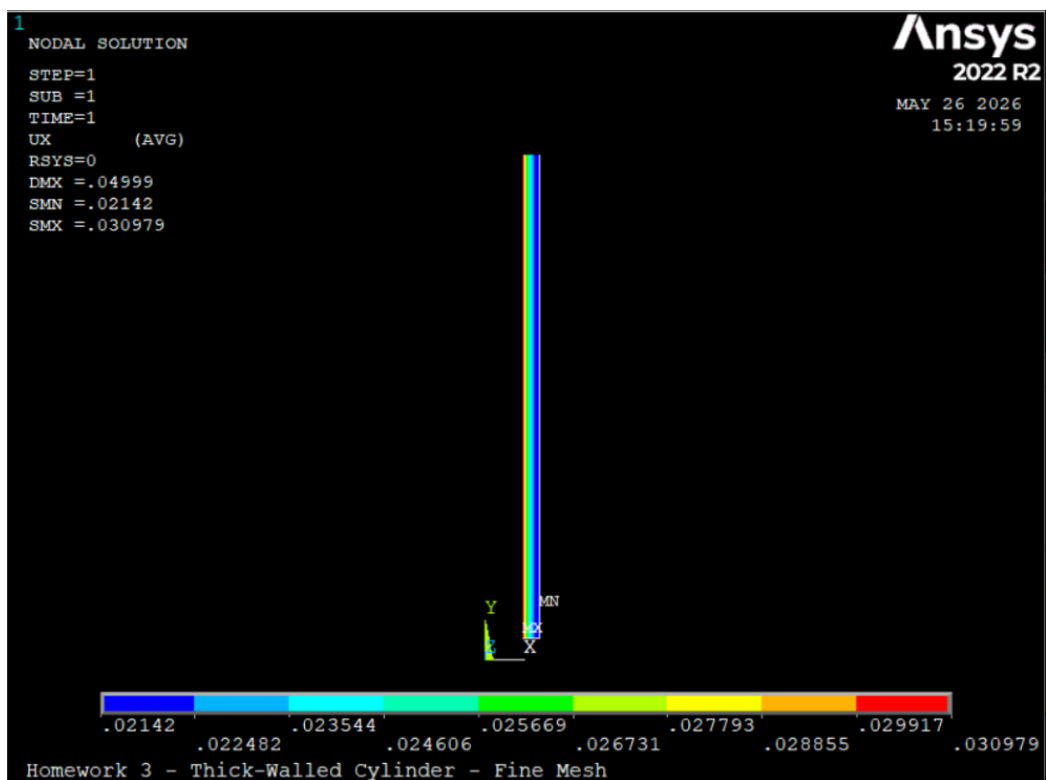


Figure 3 Radial displacement contour

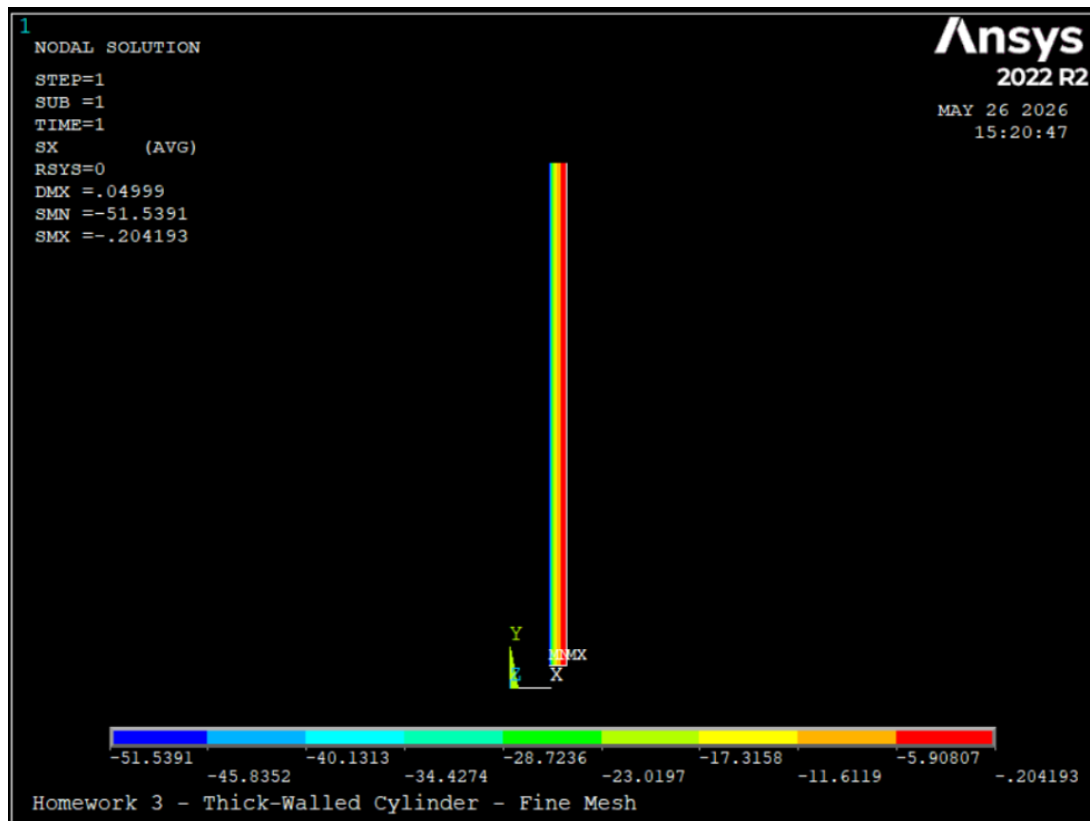


Figure 4 Radial stress contour

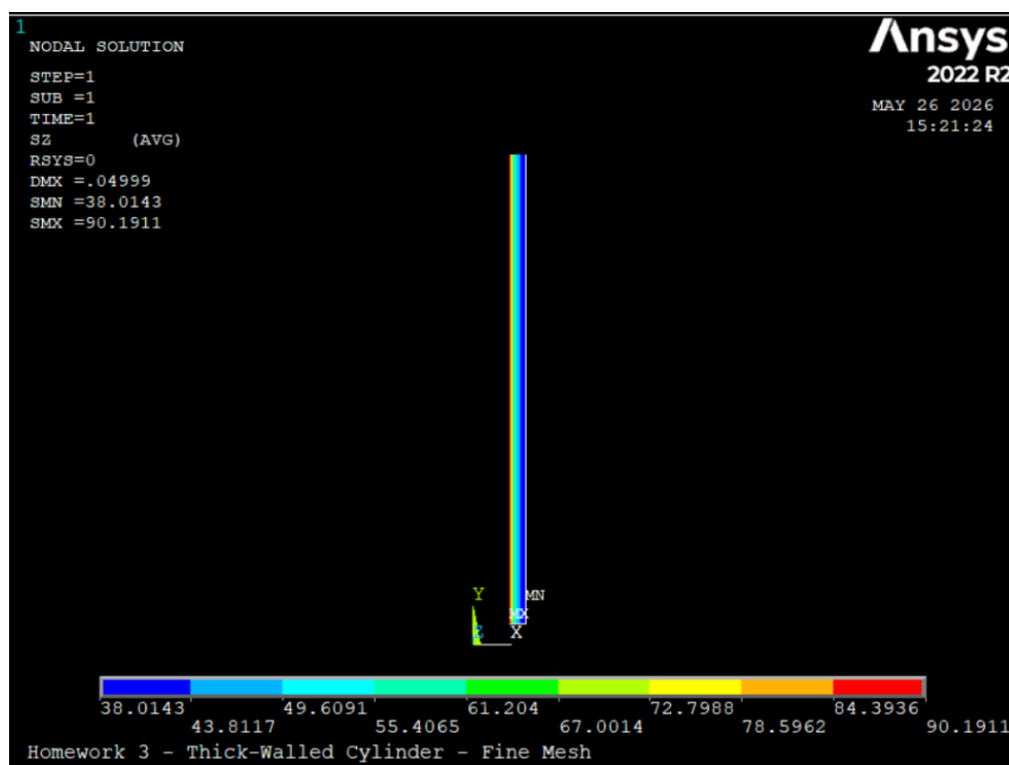


Figure 5 Fine mesh hoop stress contour

4. Radial Path Results at Mid-Length Section

For the fine mesh, a radial path was defined at the mid-length section:

$$z = L / 2 = 690 \text{ mm}$$

The path extends from the inner surface to the outer surface:

$$r = 58 \text{ mm} \rightarrow 113 \text{ mm}$$

In ANSYS, the path coordinate was printed as S, where S = 0 corresponds to the inner surface and S = 55 corresponds to the outer surface. Therefore, the physical radial coordinate is:

$$r = 58 + S$$

The path extraction results include $SIGR = SX = \sigma_r$, $SIGT = SZ = \sigma_\theta$, and $URAD = UX = u_r$. The extracted path data show that the radial stress increases from approximately -51.515 MPa at the inner surface to -0.207 MPa at the outer surface, while the hoop stress decreases from approximately 90.160 MPa to 38.016 MPa. The radial displacement decreases from approximately 0.030974 mm to 0.021421 mm along the same path.

r (mm)	FEM σ_r (MPa)	FEM σ_θ (MPa)	FEM u_r (μm)
58	-51.5150	90.1600	30.974
69	-31.9020	69.8010	27.380
80	-18.8740	56.7790	24.974
91	-10.2790	48.1880	23.327
102	-4.3125	42.2230	22.192
113	-0.2073	38.0160	21.421

Table 5. Selected fine mesh path results at $z = L / 2$

The radial stress distribution is compressive throughout the wall thickness and approaches zero at the outer surface, which is consistent with the traction-free boundary condition. The hoop stress is tensile and reaches its maximum value at the inner surface. This is also physically expected for a thick-walled cylinder subjected to internal pressure.

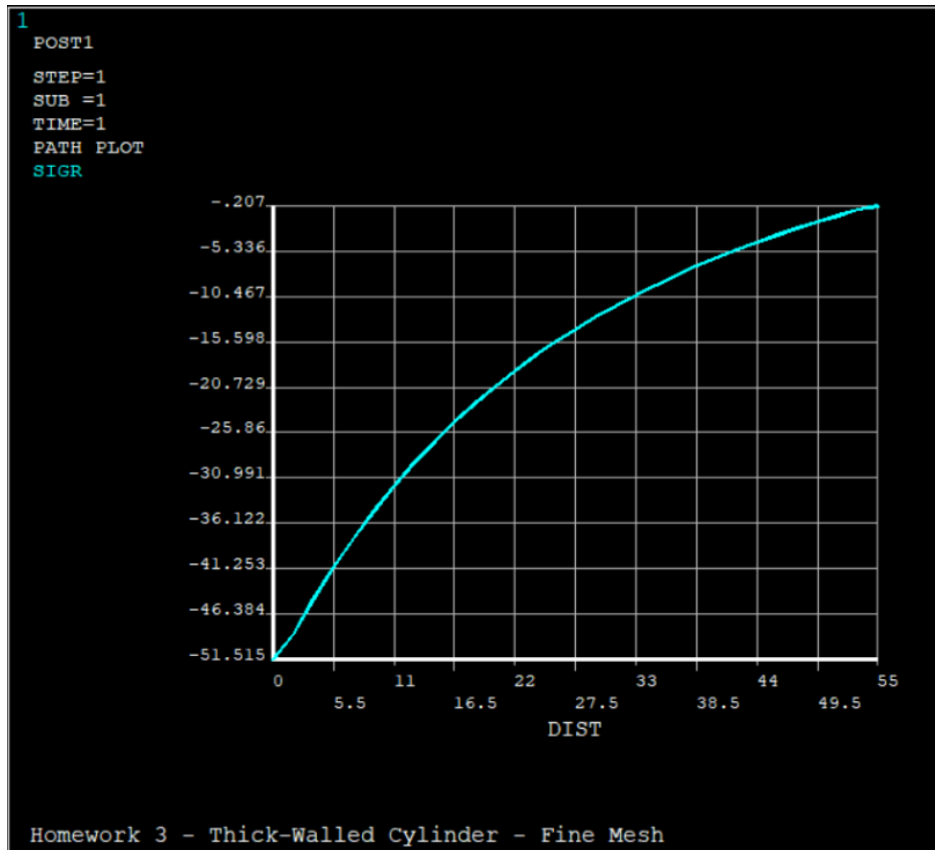


Figure 6 Radial stress path plot

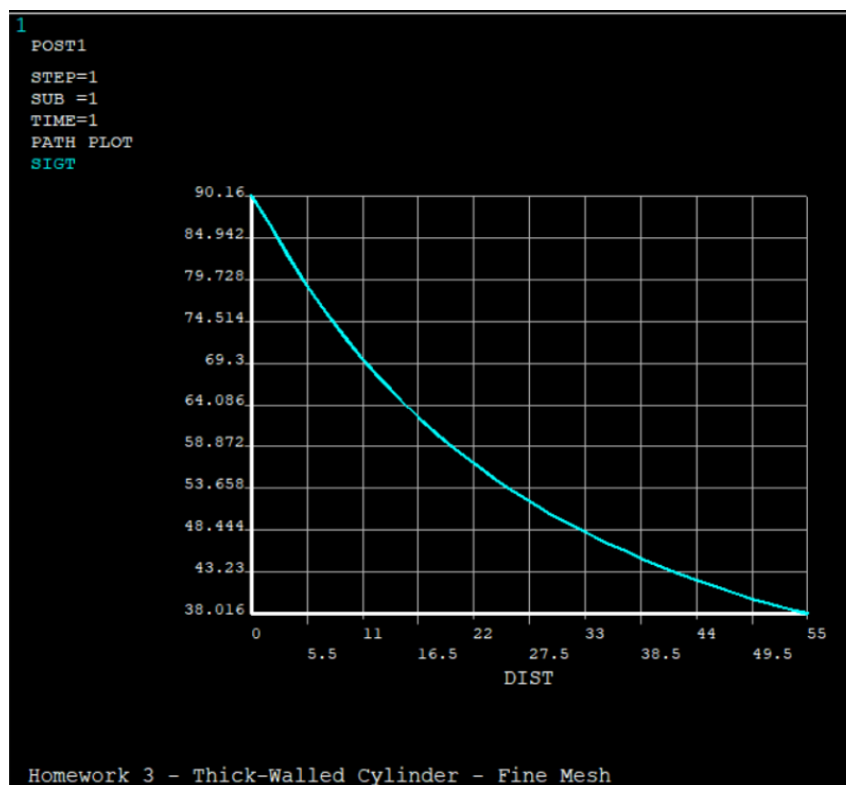


Figure 7 Hoop stress path plot

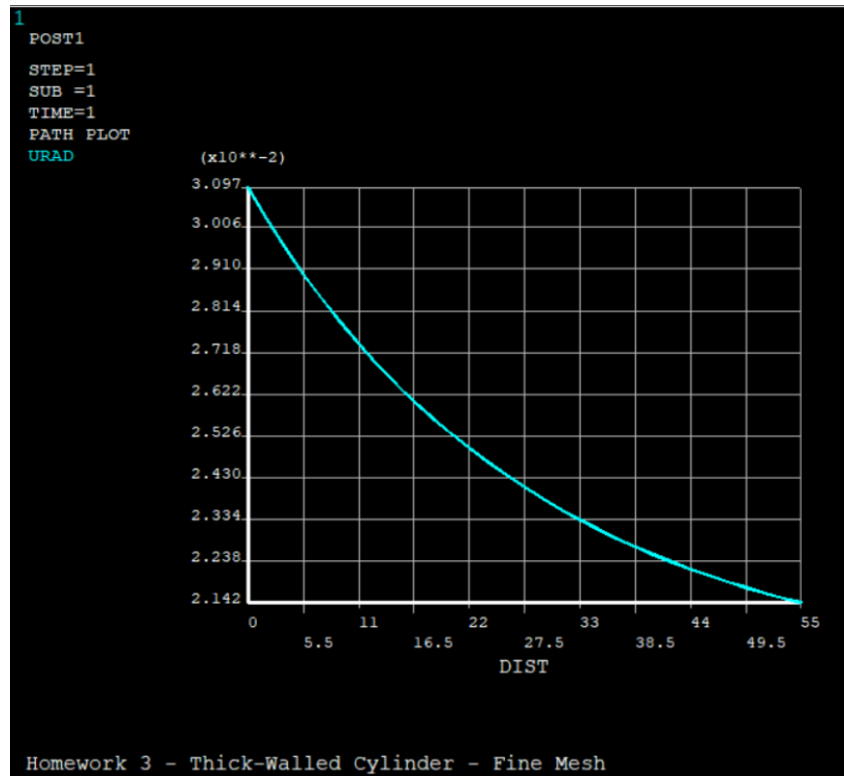


Figure 8 Radial displacement path plot

5. ANALYTICAL LAMÉ SOLUTION

The finite element results were compared with the analytical Lamé solution for a thick-walled cylinder under internal pressure. The analytical stress distributions are:

$$\sigma_r(r) = \frac{P_i r_i^2}{r_o^2 - r_i^2} \left(1 - \frac{r_o^2}{r^2} \right) \quad \sigma_\theta(r) = \frac{P_i r_i^2}{r_o^2 - r_i^2} \left(1 + \frac{r_o^2}{r^2} \right)$$

The radial displacement for the open-ended cylinder is:

$$u_r(r) = \frac{P_i r_i^2}{E(r_o^2 - r_i^2)} \left[(1 - \nu)r + \frac{(1 + \nu)r_o^2}{r} \right]$$

Using these formulas, the analytical values at the inner and outer surfaces are:

Location	σ_r (MPa)	σ_θ (MPa)	u_r (μm)
$r = r_i = 58 \text{ mm}$	-53,0000	90,9143	30,9761
$r = r_o = 113 \text{ mm}$	0,0000	37,9143	21,4216

Table 6. Analytical reference values

6. FEM Versus Analytical Comparison

The fine mesh FEM results were compared with the analytical Lamé solution along the radial path at $z = L/2$. Selected comparison values are given in Table 7.

r (mm)	FEM σ_r (MPa)	Analytical σ_r (MPa)	FEM σ_θ (MPa)	Analytical σ_θ (MPa)	FEM u_r (μm)	Analytical u_r (μm)
58	-51,5150	-53,0000	90,1600	90,9143	30,974	30,9761
69	-31,9020	-31,8859	69,8010	69,8002	27,380	27,3813
80	-18,8740	-18,8653	56,7790	56,7796	24,974	24,9757
91	-10,2790	-10,2741	48,1880	48,1884	23,327	23,3281
102	-4,3125	-4,3093	42,2230	42,2236	22,192	22,1933
113	-0,2073	0,0000	38,0160	37,9143	21,421	21,4216

Table 7. Fine mesh FEM versus analytical Lamé solution

The agreement between the fine mesh FEM results and the analytical Lamé solution is very good. The radial displacement almost exactly matches the analytical solution throughout the cylinder wall. The hoop stress distribution also shows very close agreement, with only a small difference near the inner surface.

The radial stress result shows the largest difference at the inner surface. The finite element value at the inner surface is -51.515 MPa, while the analytical value is -53 MPa. This corresponds to an error of approximately 2.8%. This discrepancy is acceptable and can be attributed mainly to stress recovery and nodal averaging in finite element post-processing. At the outer surface, the radial stress is -0.207 MPa, which is very close to the analytical traction-free value of zero.

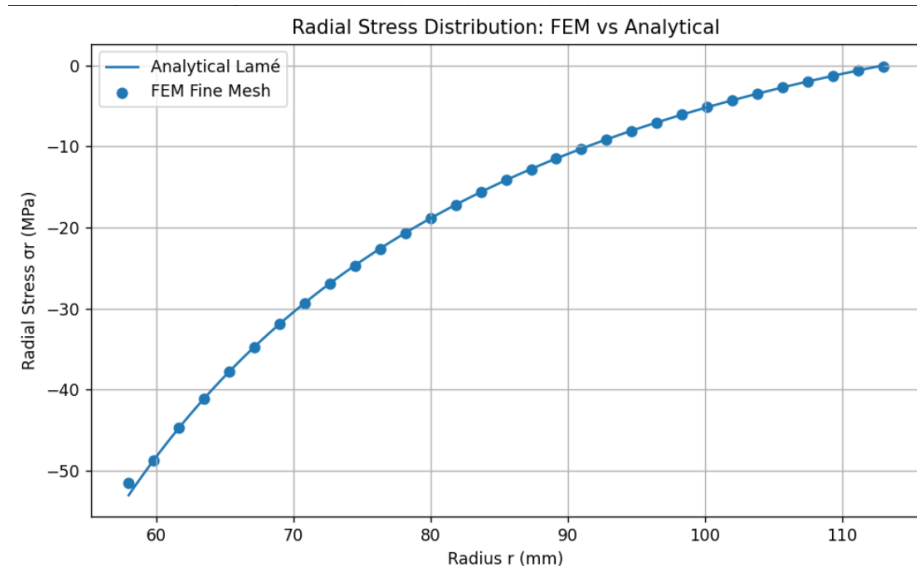


Figure 9 Radial stress comparison between fine mesh FEM and analytical Lamé solution

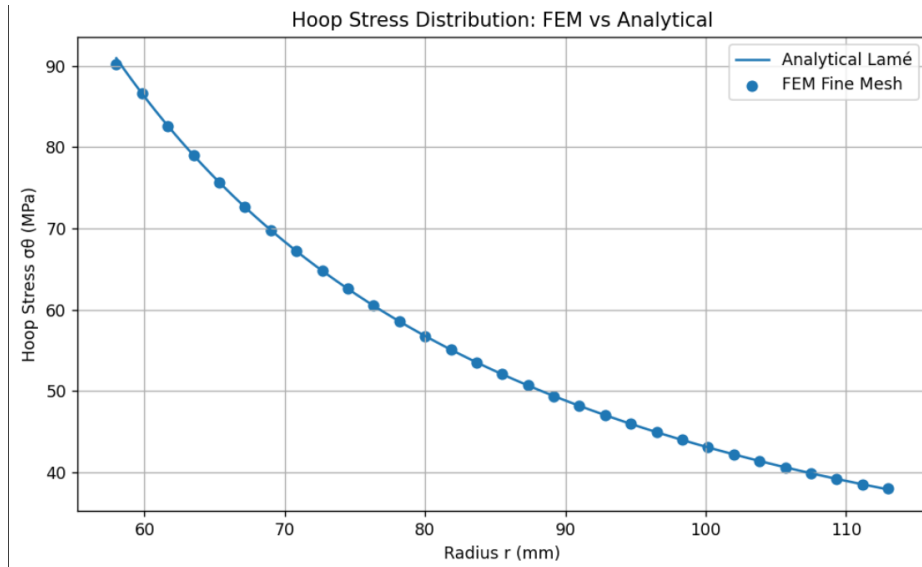


Figure 10 Hoop stress comparison between fine mesh FEM and analytical Lamé solution.

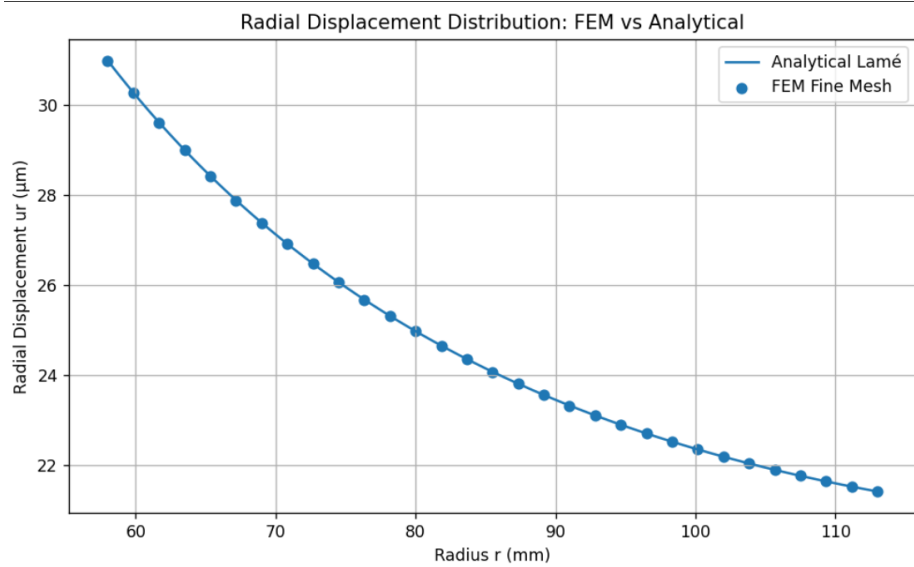
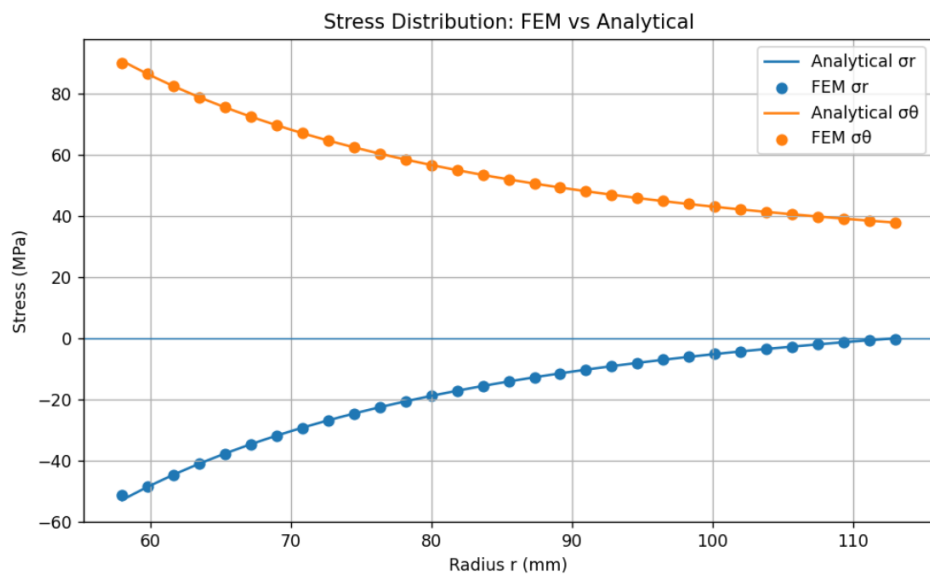


Figure 11 Radial displacement comparison between fine mesh FEM and analytical Lamé solution



7. DISCUSSION

The finite element model successfully captures the mechanical response of the thick-walled cylinder under internal pressure. The displacement results converge rapidly and agree almost exactly with the analytical solution. This is expected because displacement is the primary solution variable in the finite element method.

The stress results are more mesh-dependent because stresses are obtained from displacement gradients. As the mesh is refined, both radial and hoop stress results approach the analytical Lamé solution. The fine mesh provides sufficiently accurate results for the stress and displacement distributions.

The radial stress is compressive and increases from the inner surface toward the traction-free outer surface. The hoop stress remains tensile throughout the wall thickness and reaches its maximum value at the inner surface. This behavior is consistent with the theoretical stress distribution in internally pressurized thick-walled cylinders.

The small discrepancy in radial stress near the inner boundary is mainly due to stress averaging and stress recovery in the finite element post-processing. Since the displacement and hoop stress results are highly accurate and the radial stress follows the correct trend, the fine mesh model can be considered reliable.

8. CONCLUSION

A 2D axisymmetric finite element model of a thick-walled cylinder under internal pressure was developed using ANSYS Mechanical APDL. The model used PLANE182 axisymmetric elements and linear elastic material properties. Three progressively refined meshes were analyzed to study mesh convergence.

The radial displacement results showed excellent agreement with the analytical Lamé solution for all meshes. Stress results improved with mesh refinement, and the fine mesh gave close agreement with the analytical solution. The fine mesh path results at $z = L/2$ confirmed that the radial stress, hoop stress, and radial displacement distributions are consistent with the expected behavior of a thick-walled cylinder under internal pressure.

Therefore, the finite element model is considered validated against the analytical Lamé solution.

9. INPUT FILE

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nax=200

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FINISH

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PLNSOL, S, Z

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